A 2D/3D Vision Chip based on Organic Substrate 3D Package

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Abstract: This paper describes a 2D/3D vision chip with integrated sensing and processing capabilities. The vision chip integrates a 1920×1080 2D/3D image sensor and a visual vector processor. The image sensor generates a gray-scale image at 300 frames/s or depth maps (indirect Time-of-Flight, iToF) at 30 frames/s. The visual processor is a programmable intelligent visual vector processor with a peak energy efficiency of 1.49 TOPS/W. The vision chip achieves real-time end-to-end processing of convolutional neural networks (CNNs) and conventional image processing algorithms. Furthermore, an end-to-end 2D/3D vision system is built to exhibit the capacity of the vision chip. The vision system achieves real-timing applications under 2D and 3D scenes, such as human face detection and depth map reconstruction. The frame rate of image acquisition, image process, and result display is larger than 30 fps.

Key words: Vision chip; 2-D/3-D image processing; near-sensor computing; convolutional neural networks

1. Introduction

Vision chip integrates an image sensor and parallel processing elements (PE) array to form a compact end-to-end visual system-on-chip (SoC)[1]. Vision chips of the first-generation couple PEs and pixels to achieve maximum parallelism with ultra-low latency. This type of integration causes passive interaction of both the sensing and processing parts. The fill factor and pixel area are worsened because of PE insertion, and the functionality of PE is forced to be simplified because of the limited area for computing circuits in pixels. Recently, researchers have applied 3D stack technology on vision chips[2] to integrate sensors and visual processors. 3D stack requires a design constraint on sensors and processors that these chips need to be designed as same size for die-to-die vias to interconnect.

Nowadays, novel image sensors have expanded from single 2D data to multimodal data (2D/3D). Introducing the concept of depth maps, the processing of 2D/3D images is widespread in both industrial and recreational markets such as automotive[5], robotics[6], and augmented reality[7]. Simultaneously, intelligent visual processors also support processing corresponding 2D/3D image data. When it comes to 3D stacked 2D/3D vision chips, new

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challenges arise. Pixels for 3D sensing implement extra structure, leading to a larger pixel size and driven current than conventional 2D pixels. For example, the four-phase integration indirect time-of-flight (iToF) pixel measures the phase shift of reflection light, which requires extra modulation gates and storage diodes for separate phases. The increasing area of the pixel array makes matching the die size and pin locations of 2D/3D sensors and processors a big challenge in the co-design process, especially when the technology nodes of sensors and processors are different. This mismatch in die size makes 3D stacking a costly solution to 2D/3D vision chips and a severe constraint on the logic design of the visual processor to be integrated.

With the development of package technology, organic substrate provides high-speed interconnection for chip integration. Heterogeneous chips with different functionality, size, and technology nodes can be integrated with customized substrates[3]. By integrating the two chips through high-density and low-loss substrate technology, it is possible to achieve a vision chip[4] that combines high-quality sensing and complex computing operations simulating the visual system of the human eye that combines the retina and visual cortex. Therefore, those vision chips apply advanced 3D package technology to integrate a high-resolution image sensor and an independent image processor.

To this end, a vision chip integrating 2D/3D image sensors and intelligent processors with organic substrate technology can solve the fusion of sensing and computing at 2D/3D end-to-end scenes.

In previous work, a novel 2D/3D image sensor[8] and a 2D/3D perception visual vector processor[9] were proposed. The image sensor implements iToF technology for 3D sensing and 1920×1080 full-HD resolution and its differential pixel and readout structure is also compatible with 2D sensing. The processor is a programmable intelligent visual vector processor that can implement deep neural networks and conventional image processing algorithms through customized instruction.

This paper presents a 2D/3D vision chip and a demonstration system. Organic substrate technology is applied to integrate the above sensor and processor into the 2D/3D vision chip, which can acquire 2D/3D raw images at high speed and intelligently process those images. This vision chip achieves end-to-end multimodal sensing and processing at edge scenes. A 2D/3D vision system is built through the vision chip, an FPGA, and an HDMI screen to display real-time sensing images and processing results. Convolutional neural networks and depth map reconstruction algorithms have been implanted in the system.

The paper is organized as follows: Section 2 overviews the 2D/3D vision system architecture and algorithm implementation. Section 3 characterizes the design of the vision chip. Section 4 provides the measurement results, and Section 5 concludes this work.

2. Design of the 2D/3D Vision Chip

2.1. 2D/3D Vision chip architecture

Fig. 1 shows the 2D/3D vision chip. This vision chip integrates a 1920×1080 2D/3D image sensor[8] and a visual vector processor[9]. The sensor implemented a backside-illuminated 6 × 6 μm pinned photodiode pixel and differential multiplexing readout architecture to realize 2D (gray-scale) and 3D (four-phase integration iToF) working modes. The sensor chip is fabricated in a 110-nm backside illumination CMOS image sensor process with a low depth noise of under 0.43% over the range of 0.3–1.5 m, the 2D and 3D frame rates achieve 300 and 60 frames/s, respectively with 3μs row-time 12-bit ADCs. A clock edge detection and error compensation structure is applied in the sensor to correct missing bits from ADCs. The processor is a programmable intelligent visual vector processor compatible with multimodal 2D/3D visual data processing. The processor mainly consists of a reconfigurable vectorial processing element (PE) array, a data memory that can be accessed fine- or coarse-grained, a micro control unit (MCU), the image-preprocessing circuit to adjust raw images before PE processing, and a group of MIPI TX and RX interface for data transfer. The vectorial PE array with neighbor PE access increases the data reuse rate and parallel computation efficiency. It can implement both convolutional neural networks (CNNs) and conventional image processing algorithms by fixed-point operations. Each image is processed by rows in the PE array. Inside the PE, it has microcomputing operation circuits to support basic operations for image processing such as add, plus, minus, multiplication, comparison, logic, multiplication-accumulation, etc. The chip is fabricated in a 55-nm CMOS process. The experimental results showed that the energy efficiency and peak performance of the processor attained as high as 1.49 TOPS/W and 409.6GOPS at 200MHz, respectively. An organic substrate is applied to integrate the sensor and the processor.

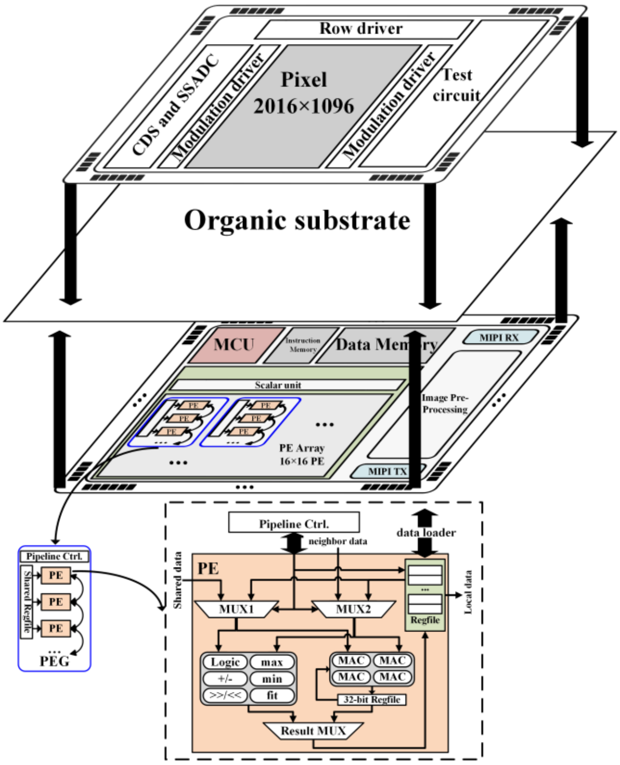


Fig. 1. 2D/3D vision chip.

2.2. Sensor and processor co-design

Considering the 3D application of the vision chip, the sensor and processor parts need specific optimizations. The reconstruction of depth maps requires the sensor to generate modulated signals from 4 phases Q(0, 0.5π, π, 1.5π). The depth d of the sensor can be calculated using the following formula, $\alpha = Q_{\pi} - Q_0$ and $\beta = Q_{1.5\pi} - Q_{0.5\pi}$, f is the modulated frequency[7]:

$$\varphi = \frac{\pi}{2} + \frac{\pi}{2}\frac{\alpha}{|\alpha| + |\beta|}, \beta \leq 0 \ or$$

$$\frac{3\pi}{2} - \frac{\pi}{2}\frac{\alpha}{|\alpha| + |\beta|}, \beta > 0$$

$$d = \frac{c}{2}\frac{\varphi}{2\pi f}$$

Those input data need a large scale of memory on the processor to restore Q(0, 0.5π, π, 1.5π). To reduce the memory, the sensor pixel and readout circuits are designed as differential structures to preprocess modulated signals and read out the intermediate results α and β. It reduces 50% of the data memory requirement on the processor. The processor data memory is designed as 4.5 MB, which is appropriate for 2 1080P modulated signals. CORDIC (coordinated rotation digital computer) is applied to implement the depth reconstruction formula, which is supported by the fixed-point operation of the PEs.

Considering the high-speed 2D application of the vision chip. The maximum frame rate of the sensor is 300 fps with a data-width of 12-bit. The data rate requirement is 7.46 Gbps. Therefore, a pair of MIPI interfaces is implanted on the sensor part and the processor part to transfer sensor data with a peak throughput of 9.6 Gbps (800 Mbps per lane). The data-width of raw images from the sensor is 12-bit. Because common fixed-point image processing implements 8-bit width or 16-bit width, those raw images are cut to 8-bit or padded to 16-bit through the image preprocessing circuit in the processor part. The total delay of the image pre-processing circuits is 0.125 ms.

2.3. 2D/3D vision system architecture

Fig. 2 shows the 2D/3D vision system, comprised of an FPGA, DDR, and a HDMI screen. The FPGA is used to configure the vision chip and to transfer the results and images from the vision chip to the HDMI screen through the AXI system bus. A MIPI CSI-2 interface is implemented in the FPGA to receive images from the vision chip. The MCU soft IP core is implemented in the MCU to transfer data through the AXI interconnect from DDR and the vision chip. The MCU also runs a language C script to overlay the result on the corresponding image and send it to the HDMI screen. For example, the result of the human face detection algorithm based on CNN is the (x, y) coordinates and (width, height) of the detected box of a human face from the raw image, the FPGA controller draws boxes on the screen with (x, y) and (width, height) to show the location of the human face in the screen when receiving the message of processing finished from the vision chip.

Fig. 3 shows the workflow of the vision system. Firstly, the FPGA reads out the instructions of the vision chip processor from DDR and transmits it to the vision chip through the QSPI interface and configure the vision chip sensor as 2D or 3D mode by IIC interface, then the independent work of the vision chip begins. The vision chip sensor continuously reads out raw images to FPGA and the vision chip processor executes the algorithm instructions to infer the input image from the sensor. The vision chip sends out processing results and messages to the FPGA transferring raw images and processing results to the HDMI screen.

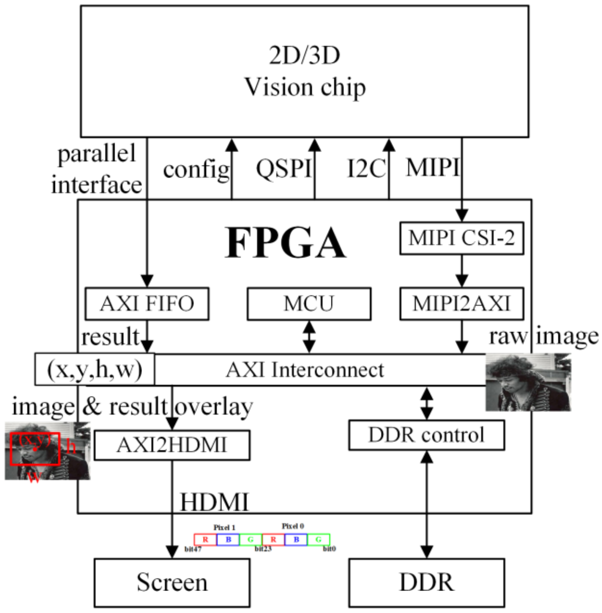


Fig. 2. Architecture of 2D/3D vision system.

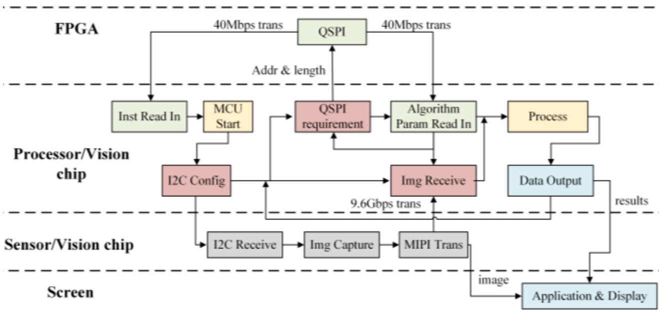


Fig. 3. 2D/3D vision system workflow.

2.2. Vision chip 3D architecture design

Fig. 4 shows the 3D architecture in a side view, bottom view, and top view. The sensor and the processor are packaged on the organic substrate. The MIPI transmitter and receiver are put on the same side to reduce the loss of high-speed signal transmission. The sensor on the top side is integrated by wire bonding, the bonding wire is covered by epoxy. The processor on the bottom is a flip chip. Two pieces of copper are exposed on the bottom as the radiator region. The insertion loss between the MIPI negative and positive lane differential ports is less than -30 dB at 800 Mbps, and the echo loss is greater than -0.035 dB at 800 Mbps The inter-channel crosstalk is less than -80 dB.

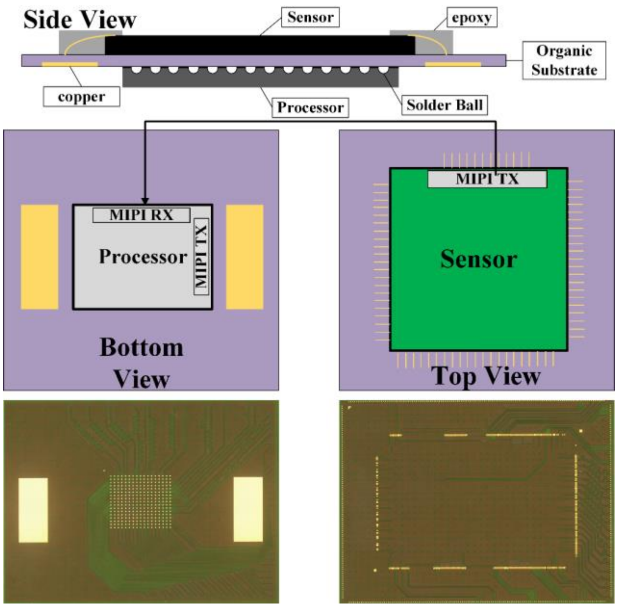


Fig. 4. 2D/3D vision chip 3D architecture.

3. Algorithm implementation

The instruction set architecture of this vision chip is a fine-grained instruction set, which is used to confirm micro computing operators. The neural network architecture applied in this work includes convolutional layers, fully-connected layers, and matrix-multiplication. 8-bit NN quantization is a proven data width that has less accuracy loss compared with the float-point model. The NN quantization method we applied on this vision chip is described in[10] [11] and the tools to simulate the accuracy loss of 8-bit parameters are described in[12]. After we get 8-bit parameters, the process of mapping from the float-point model to the fixed-point on-chip model has two steps. The first step is to transfer float-point operators of the CPU/GPU framework to the instructions of the vision chip. We packaged some common operators such as convolution to replace float-point operators in the assembly code of our vision chip. The second step is to reshape the 8-bit parameters as an efficient type for the vision chip memory access.

As depicted in Fig. 5, each layer of a neural network is separated into vectorial dataflow and performed by rows parallelly. After accumulate-multiplication, the partial sum (P-sum) is quantized as 8-bit integers with the scale factor of mul/2^shift. An activation function is applied to the quantized results to obtain the output feature map. Before the computation of the output feature map of a layer is finished, the intermediate result is restored in local registers of PEs to decrease memory access. Thus, an output stationary dataflow is implemented in the vision chip to process NN inference. The iterations of the output channel (Cout), input channel (Cin), and slide windows (M, N) are executed inside PE. The iteration of w is mapped to the row direction of the PE array. This fixed-point NN inference process is efficient on the proposed vision chip. Fig. 6 shows the fixed-point inference results of images from open-source datasets on the vision chip.

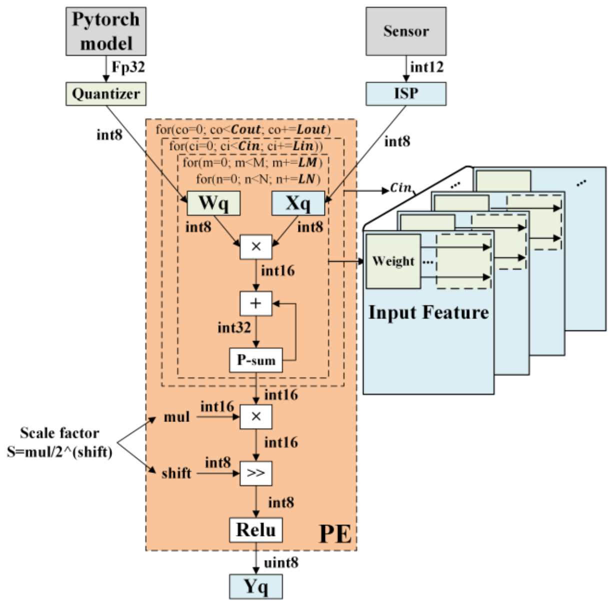


Fig. 5. Vision chip fixed-point inference of NNs.

Fig. 6. NN Inference results on the processor.

4. Measurement results

4.1. Characteristics

The processor part of the vision chip was fabricated in a 55 nm CMOS process with an area of 58.29 mm^2, the sensor part was fabricated in a 110 nm CMOS image sensor process with an area of 345.79 mm^2. The peak 8-bit performance during processing is 409.6 GOPS at 200 MHz. An iToF figure of merit (FoM_{iToF})[13] is used to estimate power consumption, depth noise, frame rate, and pixels of 3D sensors. The sensor part achieves a FoM_{iToF} of 13.7 nJ/pixel. The 2D/3D vision system is shown in Fig. 7, the vision chip is installed under a lens. The wire-bonded sensor and the processor are shown in the top view and bottom view, while the sensor bonding wires are covered by epoxy for protection. The FPGA in the system is xcvu9p-flga2104, and the screen is 1920×1080 30Hz. If 3D mode is chosen, an extra infrared light emitter board needs to be installed on the system.

We implemented a 2D human face detection NN algorithm and a 3D depth reconstruction algorithm on this system. The timing chart of human face detection and depth map reconstruction is shown in Fig. 8, and the split delay of the vision chip process including image acquisition and image processing is shown. The human face detection NN is a 67-layer model with 3 branches to differentiate the scale of human faces. The branches in the model comprise convolution, depth-wise, point-wise, and residual blocks. The processing delay of face detection is 10.2ms and the processing delay of depth map reconstruction is 4.1ms. The test environment of the 2D/3D system is shown in Fig. 9, depth of the plaster model and boxes of human faces are shown on the screen.

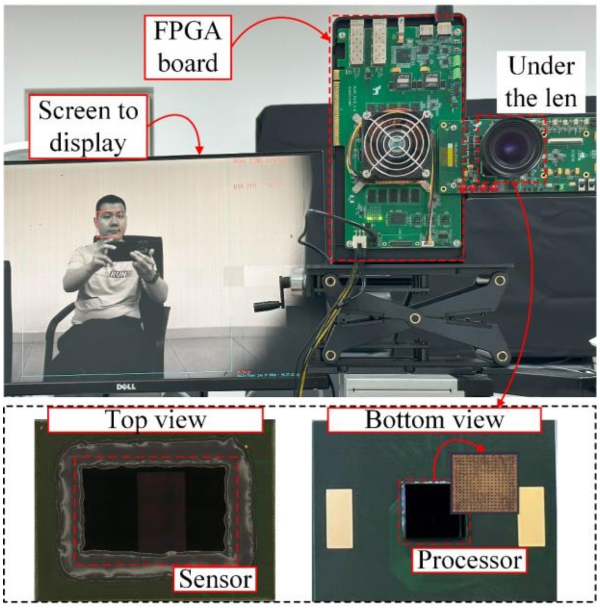
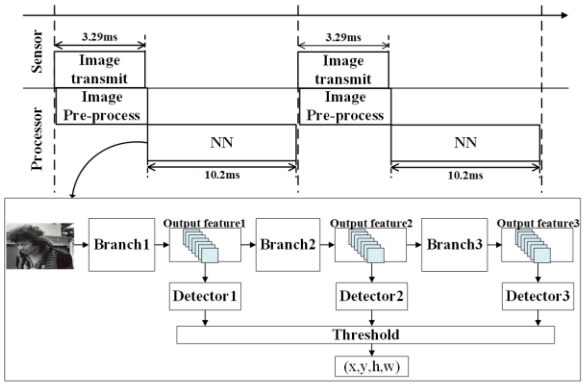
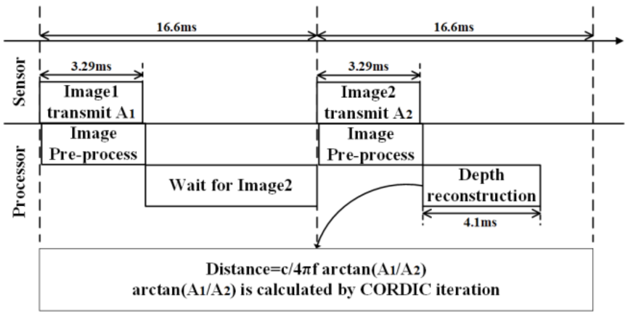


Fig. 7. The 2D/3D vision chip in the system.



(a)



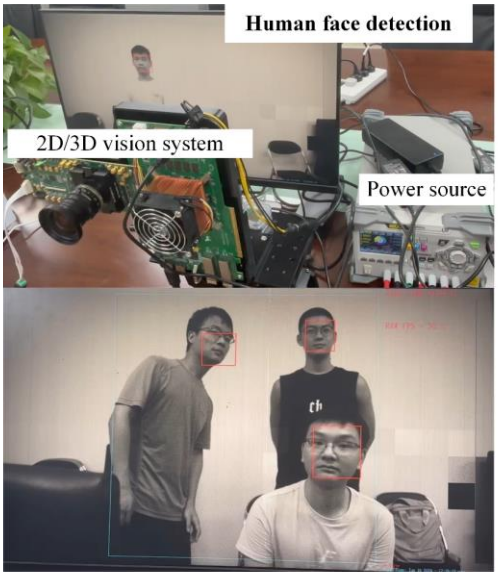
(b)

Table 1. Comparison with other NN vision chip

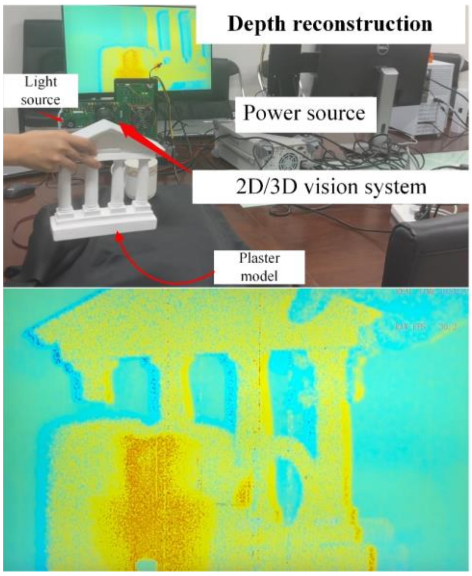
	Sony ISSCC2021 [14]	Sensors 2023 [15]	TCAS-I 2023 [16]	**This work**
Process (nm) Sensor/Processor	65/22	130	40/22	110/55
Max Clock frequency (MHz)	262.5	80	100	200
Processing Data type (int)	8b, 16b, 32b	4b	8b	8b, 16b, 32b (activation)
Imager resolution (H/V)	**4056×3040**	1024×768	1296×720	1920×1080
Peak Performance (8b)	1.21TOPS	61GOPS	**9.05TOPS**	409.6GOPS
Algorithms Support	CNN	3-layer CNN	Temporal frame filter	**2D&3D CV/CNN**
Imager data modal	RGB	Gray	RGB	**iTOF/Gray**
2D frame rate	120(2028×1520)	340	30	300
3D frame rate	-	-	-	30

Fig. 8. Timing chart. (a) Human face detection. (b) Depth map reconstruction.

Fig. 9. Test environment. (a) Human face detection. (b) Depth map reconstruction.



(a)



(b)

4.2. Comparison

Table. 1 shows a comparison of the vision chip with other NN vision chips. Among those vision chips, the proposed vision chip achieves 2D/3D multimodal sensing and end-to-end processing through instruction-level programmability and corresponding implementation methods. The programmable PE array of our processor integrates hybrid compute units to support micro-operations of multimodal algorithms. The peak performance of the proposed vision chip is not outstanding, because we use a relatively backward technology in the table and our vision chip has a small number of processing units.

The vision chip of [14] is a vision chip integrating a high-resolution image sensor and a CNN processor sub-system. It achieves end-to-end 2D RGB CNN processing with advanced technology node, but it cannot sense and process 3D images. The work of [15] is a pixel-level parallel 2D vision chip with low latency, yet it has relatively low performance when processing deep NN. The work of [16] is a vision chip for temporal frame filtering before autonomous driving processing. It achieves the highest performance, yet the application of this vision chip is specific and it needs another processor to achieve end-to-end processing.

5. Conclusion

This paper proposes a 2D/3D vision chip and corresponding sensing and processing vision system, integrating a 1080P 2D/3D sensor and a visual processor through 3D package on an organic substrate. Both 2D and 3D imaging can be obtained and real-time processed. The sensor part of our vision chip was fabricated in a 110 nm CIS process, the processor part was fabricated in a 55 nm process. The frame rate of this vision chip is 300 (2D) and 30 (3D). The peak performance of this vision chip is 409.6 GOPS. Meanwhile, we realized a real-time 2D/3D vision system based on the proposed vision chip and FPGA display system. Our vision chip is expected to promote further development of multimodal sensing and processing.

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